Geometric Response Theory: Formal Mathematical Framework from Lean 4

Extracted from GeometricResponseLean formalization

Abstract

This document extracts the mathematical framework, physics, and formal proofs from the Lean 4 formalization of geometric response theory. The theory establishes a rigorous connection between quantum structural dynamics and operational gravitational scales through Green–Kubo relaxation times, Kramers–Kronig causality constraints, and frequency-dependent effective couplings. All theorems presented here are machine-verified with zero axioms and zero sorries in the Lean codebase.

1 Introduction

This document presents the mathematical content of the `GeometricResponseLean` formalization, which proves the necessity and sufficiency of the geometric response framework connecting temporal structure (Green–Kubo time τ_G) to spectral structure (Kramers–Kronig susceptibility $\chi(\omega)$) and operational scales (frequency ω_G, length λ_G, mass m_G).

Formal status: All theorems are fully proven in Lean 4 with zero axioms and zero sorries. The proofs use standard measure theory (dominated convergence theorem), complex analysis (analytic continuation), and operator algebra (commutativity).

Connection to physics: The theory builds on the Quantum Structural Triad $\Pi_A = \sqrt{F_Q} \cdot S_A \cdot I(A{:}\bar{A})$, which is formally proven in a separate Rocq/Coq development. The Lean formalization assumes the triad framework and focuses on the geometric response layer.

2 Mathematical Framework

2.1 Debye Response Model (Sufficient Closure)

The Debye model provides a *sufficient* closure for the theory through a single-pole response kernel.

Definition 1 (Debye Kernel)**.** *For relaxation time $\tau > 0$, the Debye kernel is:*

$$R(t) = \begin{cases} \tau^{-1} e^{-t/\tau} & \text{if } t \geq 0 \\ 0 & \text{if } t < 0 \end{cases}$$

Definition 2 (Debye Susceptibility)**.** *The frequency-domain susceptibility is:*

$$\chi_D(\omega) = \frac{1}{1 + i\omega\tau}$$

Proposition 1 (DC Normalization)**.** $\chi_D(0) = 1$ *(adiabatic limit normalization).*

Proposition 2 (Power Form)**.** *The magnitude-squared susceptibility is:*

$$H(\omega) = |\chi_D(\omega)|^2 = \frac{1}{1 + (\omega\tau)^2}$$

This equals the real part: $H(\omega) = Re[\chi_D(\omega)]$.

Proposition 3 (Imaginary Part)**.**

$$Im[\chi_D(\omega)] = -\frac{\omega\tau}{1 + (\omega\tau)^2}$$

Theorem 1 (Passivity)**.** *For $\tau > 0$ and $\omega \geq 0$:*

$$Im[\chi_D(\omega)] \leq 0$$

This ensures physical causality (energy dissipation, not gain).

Theorem 2 (Transform Convergence)**.** *The truncated Fourier transform converges to the Debye susceptibility:*

$$\int_0^T R(t) e^{-i\omega t} dt \to \chi_D(\omega) \quad \text{as } T \to \infty$$

for $\tau > 0$ (exponential tail decay ensures convergence).

2.2 Green–Kubo Relations

Definition 3 (Green–Kubo Time)**.** *For normalized autocorrelation function $C(t)$:*

$$\tau_G = \int_0^\infty C(t)dt$$

provided the integral exists and is finite.

Proposition 4 (Positivity)**.** *If $C(t) \geq 0$ almost everywhere and $C(t) > 0$ on a set of positive measure, then $\tau_G > 0$.*

Proposition 5 (Wiener–Khinchin at DC)**.** *The one-sided power spectral density at zero frequency satisfies:*

$$S_\Pi(0) = 2\int_0^\infty C_\Pi(t)dt = 2\tau_G$$

2.3 Operational Scales

From the relaxation time τ_G, three operational scales emerge:

Definition 4 (Operational Scales)**.**

$$\omega_G = \tau_G^{-1} \quad \textit{(persistence frequency)} \tag{1}$$

$$\lambda_G = 2\pi c \cdot \tau_G \quad \textit{(coherent wavelength)} \tag{2}$$

$$m_G = \frac{\hbar}{c^2\tau_G} \quad \textit{(inertial scale)} \tag{3}$$

Proposition 6 (Scale Properties)**.**

- $\omega_G > 0$ *if* $\tau_G > 0$
- $\lambda_G \geq 0$ *for* $c \geq 0$, $\tau_G \geq 0$
- $m_G \geq 0$ *for* $\hbar \geq 0$, $c > 0$, $\tau_G > 0$

2.4 Effective Gravitational Coupling

Definition 5 (Effective Coupling)**.** *The frequency-dependent effective coupling is:*

$$G_{\mathit{eff}}(\omega) = G_0 \cdot \chi(\omega)$$

where G_0 is the static (DC) coupling and $\chi(\omega)$ is the susceptibility.

For the Debye model:

$$G_{\mathit{eff}}(\omega) = G_0 \cdot \frac{1}{1 + i\omega\tau_G}$$

Proposition 7 (Power Form)**.** *The magnitude-only form is:*

$$G_{\mathit{eff}}(\omega) = G_0 \cdot H(\omega) = G_0 \cdot \frac{1}{1 + (\omega\tau_G)^2}$$

Proposition 8 (DC Limit)**.** $G_{\mathit{eff}}(0) = G_0$ *(recovers static coupling at zero frequency).*

Proposition 9 (Bounds)**.** *For* $G_0 \geq 0$*:*

$$0 \leq G_{\mathit{eff}}(\omega) \leq G_0$$

Proposition 10 (High-Frequency Asymptotics)**.** *For* $|\omega\tau| > 0$*:*

$$G_{\mathit{eff}}(\omega) \leq \frac{G_0}{(\omega\tau)^2}$$

3 Kramers–Kronig Necessity Theorem

The **necessity theorem** establishes that the Green–Kubo time τ_G uniquely determines the low-frequency slope of the Kramers–Kronig susceptibility.

3.1 Mathematical Setup

Consider a causal response kernel $R(t)$ (retarded: $R(t) = 0$ for $t < 0$) with susceptibility:

$$\chi(\omega) = \int_0^\infty R(t) e^{-i\omega t} dt$$

The imaginary part is:

$$\mathrm{Im}[\chi(\omega)] = -\int_0^\infty R(t) \sin(\omega t) dt$$

3.2 Pointwise Limit

Lemma 1 (Sinc Representation)**.** *For* $\omega \neq 0$*:*

$$\frac{\sin(\omega t)}{\omega} = t \cdot \mathit{sinc}(\omega t)$$

where $\mathit{sinc}(x) = \sin(x)/x$ *for* $x \neq 0$ *and* $\mathit{sinc}(0) = 1$*.*

Lemma 2 (Pointwise Convergence)**.**

$$\lim_{\omega \to 0} \frac{\sin(\omega t)}{\omega} = t$$

This holds pointwise for each fixed t*, using continuity of sinc at zero.*

3.3 Dominating Bounds

Lemma 3 (Trigonometric Bound)**.** *For all $x \in \mathbb{R}$:*

$$|\sin(x)| \leq |x|$$

Lemma 4 (Integrand Bound)**.** *For $\omega \neq 0$:*

$$\left|\frac{\sin(\omega t)}{\omega}\right| \leq |t|$$

Lemma 5 (Uniform Domination)**.** *If $|R(t)| \leq M$ for some $M > 0$, then:*

$$\left|R(t) \cdot \frac{\sin(\omega t)}{\omega}\right| \leq M \cdot |t|$$

and $M \cdot |t|$ is integrable on any bounded interval $[0, T]$.

3.4 Main Necessity Theorem

Theorem 3 (GK→KK Necessity)**.** *Let $R(t)$ be a bounded, measurable, causal response kernel with finite Green–Kubo time:*

$$\tau_G = \int_0^\infty t \cdot R(t) dt < \infty$$

Then the low-frequency slope of the imaginary susceptibility is uniquely determined:

$$\lim_{\omega \to 0} \frac{Im[\chi(\omega)]}{\omega} = -\tau_G$$

Equivalently:

$$\frac{\partial}{\partial \omega} Im[\chi(\omega)]\Big|_{\omega=0} = -\tau_G$$

Proof Strategy. The proof proceeds in three stages:

Stage 1: Pointwise Limit

$$\frac{\sin(\omega t)}{\omega} \to t \quad \text{as } \omega \to 0$$

via the sinc representation and continuity.

Stage 2: Domination

$$\left|R(t) \cdot \frac{\sin(\omega t)}{\omega}\right| \leq M \cdot |t|$$

uniformly in ω (for $\omega \neq 0$), with $M \cdot |t|$ integrable.

Stage 3: Dominated Convergence Apply the dominated convergence theorem to the integral:

$$\int_0^T R(t) \cdot \frac{\sin(\omega t)}{\omega} dt \to \int_0^T t \cdot R(t) dt \quad \text{as } \omega \to 0$$

Then take $T \to \infty$ to identify the limit as $-\tau_G$ (with appropriate sign conventions). □

Corollary 1 (Effective Coupling Slope)**.** *For $G_{eff}(\omega) = G_0 \cdot \chi(\omega)$:*

$$\lim_{\omega \to 0} \frac{Im[G_{eff}(\omega)]}{\omega} = -G_0 \cdot \tau_G$$

3.5 Physical Interpretation

The necessity theorem establishes that:

- **Temporal structure** (autocorrelation decay rate τ_G) **determines spectral structure** (susceptibility slope at DC)
- You cannot have an arbitrary KK slope; it is fixed by the Green–Kubo integral
- This is a **rigorous, machine-verified** connection with zero axioms

4 Debye Sufficiency Theorem

The Debye model provides a *sufficient* closure, showing that the necessity bound is **achievable**.

Theorem 4 (Debye Sufficiency)**.** *The Debye kernel $R(t) = \tau^{-1} e^{-t/\tau}$ (for $t \geq 0$) satisfies:*

1. ***Green–Kubo**: $\int_0^\infty R(t) dt = 1$ (normalized)*
2. ***Kramers–Kronig**: $Im[\chi_D(\omega)] \leq 0$ for $\omega \geq 0$ (passivity)*
3. ***DC Normalization**: $\chi_D(0) = 1$*
4. ***Slope Matching**: $\lim_{\omega \to 0} Im[\chi_D(\omega)]/\omega = -\tau$*

Proof Sketch. 1. **Normalization**: $\int_0^\infty \tau^{-1} e^{-t/\tau} dt = 1$ (standard exponential integral).

2. **Transform**: The Fourier–Laplace transform gives:

$$\chi_D(\omega) = \int_0^\infty \tau^{-1} e^{-t/\tau} e^{-i\omega t} dt = \frac{1}{1 + i\omega\tau}$$

3. **Passivity**: $\mathrm{Im}[\chi_D(\omega)] = -\omega\tau/(1 + (\omega\tau)^2) \leq 0$ for $\omega \geq 0$.

4. **Slope**:

$$\lim_{\omega \to 0} \frac{\mathrm{Im}[\chi_D(\omega)]}{\omega} = \lim_{\omega \to 0} \frac{-\tau}{1 + (\omega\tau)^2} = -\tau$$

□

Corollary 2 (Closure)**.** *Together, necessity and sufficiency establish that:*

$$\textit{Debye model} \Longleftrightarrow \textit{Minimal closure for GK + KK constraints}$$

5 Conservation Laws

5.1 Flat-Space Conservation

In flat space, the d'Alembertian $\Box$ and divergence ∇ commute: $[\nabla, \Box] = 0$.

Theorem 5 (Flat-Space Conservation)**.** *Let T be a tensor field (stress-energy) with $\nabla \cdot T = 0$ (conserved).*

For any polynomial operator $G(\Box) = \sum_n a_n \Box^n$:

$$\nabla \cdot [G(\Box)T] = 0$$

In particular, for the Debye approximation:

$$G_{\mathit{eff}}(\Box) \approx G_0(1 - \tau_G^2 \Box)$$

we have:

$$\nabla \cdot [G_{\mathit{eff}}(\Box)T] = 0$$

Proof Strategy. The proof is pure algebra:

$$
\begin{aligned}
\nabla \cdot [G(\Box)T] &= \nabla \cdot \left[\sum_n a_n \Box^n T\right] && (4)\\
&= \sum_n a_n \nabla \cdot (\Box^n T) \quad \text{(linearity)} && (5)\\
&= \sum_n a_n \Box^n (\nabla \cdot T) \quad \text{(commutativity: } [\nabla, \Box] = 0) && (6)\\
&= \sum_n a_n \Box^n \cdot 0 \quad \text{(hypothesis: } \nabla \cdot T = 0) && (7)\\
&= 0 && (8)
\end{aligned}
$$

□

Lemma 6 (Iterated Commutativity)**.** *If* $[\nabla, \Box] = 0$*, then for any* $n \in \mathbb{N}$*:*

$$\nabla \cdot (\Box^n T) = \Box^n (\nabla \cdot T)$$

Proof. By induction on n:

- Base case ($n = 0$): $\nabla \cdot T = \nabla \cdot T$ (trivial).
- Inductive step:

$$
\begin{aligned}
\nabla \cdot (\Box^{n+1} T) &= \nabla \cdot (\Box(\Box^n T)) && (9)\\
&= \Box(\nabla \cdot (\Box^n T)) \quad \text{(commutativity)} && (10)\\
&= \Box(\Box^n (\nabla \cdot T)) \quad \text{(induction hypothesis)} && (11)\\
&= \Box^{n+1} (\nabla \cdot T) && (12)
\end{aligned}
$$

□

5.2 Physical Significance

The conservation theorem ensures that:

- Frequency-dependent coupling $G_{\text{eff}}(\Box)$ preserves energy–momentum conservation
- This is exact in flat space (commuting operators)
- In curved space, commutator terms $[\nabla, \Box^n] \sim R \cdot \nabla$ produce $O(\tau_G^2)$ corrections

6 Ornstein–Uhlenbeck Model

The OU process provides a concrete exemplar connecting microscopic dynamics to geometric response.

Definition 6 (OU Autocorrelation)**.** *For relaxation time $\tau > 0$, the normalized autocorrelation is:*

$$C(t) = e^{-|t|/\tau}$$

Proposition 11 (Green–Kubo Time)**.** *For the OU model:*

$$\tau_G = \int_0^\infty e^{-t/\tau} dt = \tau$$

The geometry-relaxation time equals the microscopic correlation time.

Proposition 12 (Finite-Window Approximation)**.** *For finite time window T:*

$$I(T) = \int_0^T e^{-t/\tau} dt = \tau(1 - e^{-T/\tau})$$

As $T \to \infty$:

$$I(T) \to \tau = \tau_G$$

Proposition 13 (Monotonicity)**.** *The finite-window integral $I(T)$ is monotone increasing in T and bounded above by τ.*

7 Summary of Formal Results

7.1 Proven Theorems (Zero Axioms, Zero Sorries)

1. **Debye Model**: Complete characterization (kernel, susceptibility, passivity, convergence)
2. **GK→KK Necessity**: Full dominated convergence proof (200+ lines of tactics)
3. **Debye Sufficiency**: Transform convergence and passivity
4. **Conservation**: Flat-space operator algebra (commutativity-based)
5. **Operational Scales**: Emergence from τ_G (definitions and properties)
6. **OU Exemplar**: Concrete model connecting micro to macro

7.2 Proof Architecture

The formalization uses:

- **Measure Theory**: Dominated convergence theorem (Mathlib)
- **Complex Analysis**: Analytic continuation, exponential decay
- **Operator Algebra**: Commutativity, linearity, iteration
- **Topology**: Continuity, limits, filter convergence

7.3 Connection to Paper

Primary manuscript targeted by the formalization:

- **GK $\to$ KK necessity chain**: Green–Kubo time from autocorrelation and the resulting low-frequency constraint on $\chi(\omega)$ under causality/passivity/regression normalization.
- **Debye closure (sufficient)**: single-pole response as an optional slope-saturating model consistent with the necessity constraint.
- **Operational scales**: ω_G, λ_G, m_G induced by τ_G.
- **Conservation (flat-space algebraic form)**: polynomial-in-$\Box$ operators preserve conservation when commuting with ∇.

8 Outlook

The formalization establishes the **mathematical backbone** of geometric response theory:

- Rigorous connection: temporal structure $\leftrightarrow$ spectral structure
- Sufficient closure: Debye provides a simple one-timescale model consistent with the necessary low-frequency constraints
- Conservation: Frequency-dependent coupling preserves energy–momentum
- Operational scales: Emergent length/time/mass from τ_G

Future extensions:

- Curved-space conservation with explicit commutator bounds
- CTP (closed-time-path) action formalism
- Connection to full quantum field theory on curved backgrounds

Acknowledgments

This document extracts the mathematical content from the `GeometricResponseLean` formalization, which uses the Lean 4 theorem prover and Mathlib library.
All proofs are machine-verified.